SEPARATEDNESS DESCENDS TO A FINITE LEVEL VIA THE DIAGONAL

THE STACKS PROJECT AUTHORS

Editorial note. No omissions: the reduction to the diagonal is explicit and the permissibility citations are given.

Allowed standard prerequisites (editorial).

- Diagonal immersion. Schemes, Lemma schemes-lemma-diagonal-immersion
- Descend closed immersion fp. Limits, Lemma lemma-descend-closed-immersion-finite-presentation (same chapter)

Source reference. Original reference 081D: sources/stacks-chow/tags.txt, source lines 10213–10213. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 01ZP: sources/stacks-chow/tags.txt, source lines 2602–2602. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01KJ: sources/stacks-chow/tags.txt, source lines 2052–2052. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01T5: sources/stacks-chow/tags.txt, source lines 2362–2362. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0816: sources/stacks-chow/tags.txt, source lines 10206–10206. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Notation and assumptions as in Situation 081D. If f is separated, then f_i is separated for some $i \geq 0$.*

Proof. Apply Lemma 01ZP to the diagonal morphism $\Delta_{X_0/S_0} : X_0 \to X_0 \times_{S_0} X_0$. (This is permissible as diagonal morphisms are locally of finite type and the fibre product $X_0 \times_{S_0} X_0$ is quasi-compact and quasi-separated, see Schemes, Lemma 01KJ, Morphisms, Lemma 01T5, and Schemes, Remark 0816. □

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